

Unit 13, Lesson 9:

Classifying Quadrilaterals

– Part 2



Benchmarks

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.



Learning Targets

- ☐ I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.
- ☐ I can use coordinate geometry to solve geometric problems involving quadrilaterals in a mathematical or real-world context.



13.9.1 Warm-Up: Between Two Numbers

Sigsbee Deep	Britton Hill	Panorama Tower
Dwight Sigsbee led the first expedition to chart the floor of the Gulf of America. The Sigsbee Deep, the deepest region of the Gulf of America, was named after him. The exact maximum depth is not known, but is estimated to be between 3,750 meters (m) and 4,384 m.	Britton Hill, located in Walton County, is the highest point in the state of Florida. Britton Hill’s elevation is 105 m.	Panorama Tower in Miami is the tallest building in Florida. It is 265 m tall.

1. Complete the inequality using the digits 0 – 9, without repeating, where B = number of Britton Hills and P = number of Panorama Towers.

$$3750 < \boxed{}\boxed{}\boxed{}\boxed{}B + \boxed{}\boxed{}\boxed{}\boxed{}P < 4384$$

2. Use a different set of digits, without repeating.

$$3750 < \boxed{}\boxed{}\boxed{}\boxed{}B + \boxed{}\boxed{}\boxed{}\boxed{}P < 4384$$



13.9.2 Collaboration: Real-World Quadrilaterals

1. Francesco is using computer software to design the layout of a house. The base of the house creates $DMHT$ with coordinates $D(-4, 1)$, $M(4, 5)$, $H(6, 1)$, and $T(-2, -3)$.
- a. Complete the table. Show your work in the table. Leave answers in exact form.

Line Segment	Length
$\overline{DM}$	
$\overline{MH}$	
$\overline{HT}$	
$\overline{TD}$	

b. Complete the table. Show your work in the table.
Leave answers in exact form.

Diagonal Line Segment	Length
$\overline{DH}$	
$\overline{MT}$	

c. Finish the statement.

$DMHT$ is classified as a _____ because

d. If 1 unit on the coordinate plane equals 5 feet (ft),
what are the dimensions of the base of the house?
Round to the nearest thousandth.

2. Jillian is determining which type of tile to place on the floor. She wants a tile in the shape of a parallelogram. The vertices of the tile options are shown.

Tile A	Tile B
$(-11, 6.5)$, $(-10, 3.5)$, $(-6, 6.5)$, and $(-7, 9.5)$	$(3, 9.5)$, $(7, 6)$, $(9, 8.5)$, and $(5, 11)$

- a. Which tile should Jillian pick if she wants a tile in the shape of a parallelogram?
- b. Explain your choice by listing the properties of parallelograms that were applied.

3. Aharon and Shreya classified $RNPF$ with vertices at $R(-3,7)$, $N(-1,7)$, $P(-3,2)$, and $F(-5,4)$. Their classifications are shown.

Aharon's Classification	Shreya's Classification
$RNPF$ is a square or rectangle because the diagonals are congruent.	Quadrilateral $RNPF$ has congruent diagonals but cannot be classified as a parallelogram.

- a. Whose classification do you agree with? Justify your answer using properties.
- b. Write a note to the person whose classification is incorrect, explaining their mistake.

4. $KVSW$ has vertices at $K(10, -4)$, $V(9, -8)$, $S(1, -4)$, and $W(6, -2)$.
- Show mathematically that $KVSW$ is a trapezoid, but not an isosceles trapezoid.
 - List the properties of an isosceles trapezoid that did not apply to $KVSW$.

5. Ariyon was graphing square $MGNR$, but mixed up her vertices. She knows that point M has coordinates $(3, 5)$ and point G has coordinates $(7, 7)$. Her two options for points N and R are shown.

Point N	Point R
$(8, 3)$	$(5, 1)$
$(9, 3)$	$(4, 1)$

- a. Which coordinates are correct for points N and R ? Justify your answer or show your work.
- b. Which properties of a square did you use to determine the correct points N and R ?



13.9.3 Your Turn!

1. $SCWK$ has vertices at $S(-3, -4)$, $C(0, -2)$, $W(0, -10)$, and $K(-3, -8)$.

$SCWK$ is classified as a(n) _____
because ...

2. Ziquan is designing a template for his 3D printer. He created two quadrilaterals using his software.

<i>TGHN</i>	<i>MWRB</i>
$T(16, 6)$, $G(18, 4)$, $H(17, -2)$, and $N(14, 4)$	$M(4, 5)$, $W(7, 4)$, $R(8, -3)$, and $B(3, 2)$

- a. Which quadrilateral can be classified as a kite?
Show your work.

- b. List the properties of a kite that you used.
- c. If 1 unit on the coordinate plane equals 6 inches (in), what are the measurements of the sides of the kite? Round to the nearest thousandth.



13.9.4 Wrap-Up: Reflection

1. Place an **X** on the line in relation to how you feel about today’s learning target.

- ☐ I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.

I need help.
I can’t get started.

I’m getting there.

I understand and have
evidence.

